PROJECT ANALYSIS

System Overview

QuitQ is a **full-stack E-Commerce platform** that enables transactions between **buyers and sellers** through an online system.

The system supports:

- Product browsing
- Shopping cart
- Secure payment
- Order processing
- Inventory management
- Seller and admin dashboards

System Actors Analysis

The system has **three primary actors**.

1.User(Customer)

Customers interact with the platform to purchase products.

2. Seller

Sellers manage their products and orders.

3. Administrator

Admin controls the entire system.

Functional Requirements Analysis

1. User Registration & Authentication
2. Product Catalog
3. Shopping Cart System
4. Checkout & Payment
5. Inventory Management
6. Order Management

Non-Functional Requirements Analysis

1. Security
2. Performance
3. Scalability

4. Reliability

System Architecture Analysis

3-Tier Architecture

1.Frontend Layer(Client)->React.js

2.Backend API->Java,Spring,JWT Auth etc

3.Database->Mysql,sql Server

Security Analysis (JWT)

1.Login

2.Server generates JWT token

3. Token payload

4. Request Authentication

5.Server Verification